

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1- Overview	2
2- Description	3
2-1 LEDs	3
2-2 Power Cable Connections	3
3- SPECIFICATIONS	4
3-1 Currents through Contactors	4
3-2 Isolation voltages	4
3-3 Contacts	4
3-4 Aux. Contact	4
3-5 Operating Coil	4
3-6 Front Panel LED's	5
3-7 Mains Input	5
3-8 Mains Fuse	5
3-9 INPUTS	5
3-10 OUTPUTS	5
3-11 Connectors Details	6
3-11-1 GP PORT	6
3-11-2 DEBUG PORT	6
3-12 ENVIRONMENT CONDITIONS	6
REVISION HISTORY	8

1- OVERVIEW

The **GRADIENT SWITCH** is intended for the MRI TwinSpeed Systems, having TWIN sets of Gradient Coils (WB, ZM).

The **GRADIENT-SWITCH** (will be called here Grad.Sw. or GSW) is a SWITCH performing 3 Poles-Double-Throw action. It is performed by 6 ON/OFF Contactors (2 per Pole).

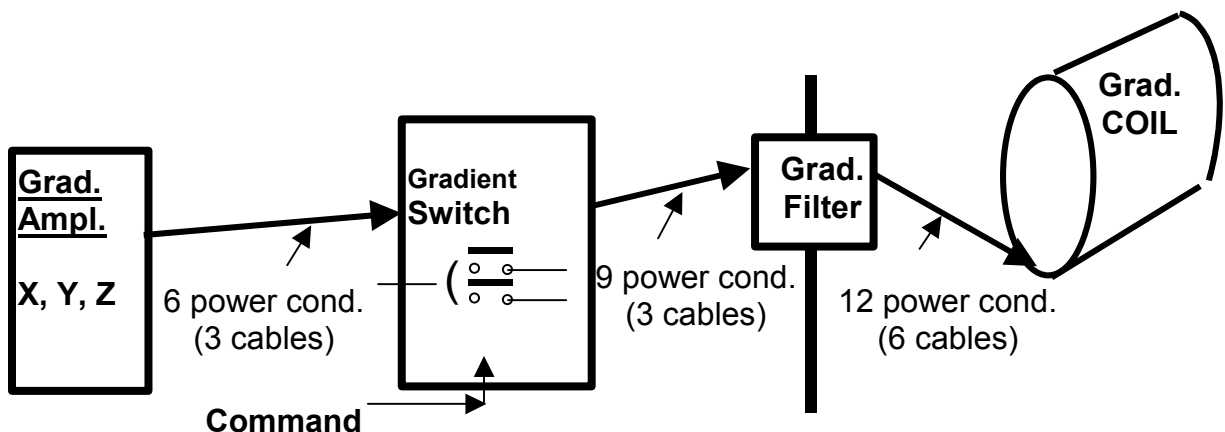
The GEMINI MRI System is composed of 2 Coils per Channel. The Channels are X, Y and Z.

The Coils are WB (Whole-Body) and ZM (Zoom). The Switch switches between the Coils WB and ZM, on the RETURN conductor, when the Main conductor is Common to both Coils.

The 3 Poles are the 3 Gradient Amplifiers and the Double-Throw is between the two Coils.

The Cables to the Gradient Coils in the Magnet, are passing through a TWIN GRADIENT FILTER on a Penetration Panel mounted on the Screen-Room wall.

The Grad.-Sw. inputs are 6 conductors from the 3 Amplifiers, and the Outputs are 9 conductors to the 6 Coils, when one conductor is common to both coils. (3 conductors per 2 Coils).



GEMINI: AMPLIFIERS TO COILS CONNECTIONS
ILLUSTRATION 1-1

The Command to the Switches is done by the Operator, through the Protocols, and via the GP for ACGD Amplifier.

2- DESCRIPTION

The **GRADIENT SWITCH** (GSW) contains mechanical Contactors in order to switch the Gradient Amplifier current from one set of Gradient Coils (WB) to other set of Gradient Coils (ZM), in the same Magnet. There are 6 contactors, 3 for the WB and 3 for the ZM.

The command to these contactors comes from the GP (situated in the ACGD Gradient Amplifier Cabinet), through RS232 cable. Other inputs/outputs, are a DEBUG to be used by a Field Engineer for testing or Trouble-shooting the GSW via his laptop or a Terminal.

A Contactor is an ON/OFF switch. The Contactors used have 3 Poles each shorted together, (parallel connected) to withstand the high current flowing. The Contactors are operated with an 120VAC supplied to its coil.

Switching ON is done by supplying 120VAC to its coil. To minimize electrical noises, the AC switching to the Contactor's coils is done at the Zero-Cross of SSR relays (Solid-State Relays).

These commands are Opto-Isolated for minimum Ground loops and minimum interferences. They also supply the Voltage isolation required by the Safety standards.

Another input is from a Terminal, PC, Laptop or the like.

Two Printed-Circuits Boards are: the GSC (Gradient Switch Controller) and the SSR (Solid State Relays). A switching PS supplies the needed voltages for the Boards.

2-1 LEDs

On the Front panel there are Led's that have the following indicators:

- 1 Green Led to show the existence of the Low voltage Power-Supply.
- 3 WB green Led's lighting when the switches are at the WB mode, 3 ZM green Led's lighting when in ZM mode.
- 3 more Yellow Led's are for ERROR. Each of them will indicate an Error in X, Y, or Z.
- [1 Yellow Led COIL TEMP – not in use]

2-2 Power Cable Connections

The Power cable connections are at the rear.

Two nuts are supplied to each stud, for secure connections. The nuts size is $\frac{3}{4}$ inch (or 19 mm). The fixed screw (STUD) protruding from the rear panel, has also $\frac{3}{4}$ inch HEX structure.

The rear connections are covered with an insulating cover with a WARNING sign for H.V.

3- SPECIFICATIONS

3-1 Currents through Contactors

The GSW Contactors will pass the high currents arriving from the gradient amplifiers to the Gradient Coils.

Contactor current is **550A RMS** for 1 Pole at 40°C, **240A RMS** for the 3 Poles simultaneously at **40°C** Ambient Temperature.

3-2 Isolation voltages

Isolation voltages:between Contactor contacts	1600 Vp
between Contacts and Coil	1000 Vp
between Live Terminals and Chassis:	1000 Vp
On SSR board :between DATA and DRIVE	2000 V.
between DATA and STATUS:	2000 V.
The cables are AWG 2 with insulation	25 kV.

3-3 Contacts

Closing Time = **40msec** max

Opening Time = **30msec** max.

Contacts MTBF at 400V/160A/Contact = **60,000** operations

Contacts heat loss at I thermal = **63** Watts

3-4 Aux. Contact

The contactors have an auxiliary contact. It is being used as a contactor status verification.

The contactor auxiliary contact sends 5 Volts DC signal when the contactor is in OPEN state, and 0Volts DC when the contactor is in CLOSE state.

3-5 Operating Coil

120 VAC , 3A , 346W start, **15W** hold.

120VAC to the contactors coil is fed through **SSR** (Solid State Relay). The SSR switches ON or OFF at the 120VAC Zero-Cross. The SSR operates by 5 Volts DC Opto-Coupler.

3-6 Front Panel LED's

NAME	COLOR	DESCRIPTION
POWER SUPPLY	Green	Indicates +5V is present on GSC Board
WB X	Green	Indicates Whole Body gradient coil for X axis is active
WB Y	Green	Indicates Whole Body gradient coil for Y axis is active
WB Z	Green	Indicates Whole Body gradient coil for Z axis is active
Zoom X	Green	Indicates Zoom gradient coil for X axis is active
Zoom Y	Green	Indicates Zoom gradient coil for Y axis is active
Zoom Z	Green	Indicates Zoom gradient coil for Z axis is active
Error X	Yellow	Indicates switch fault in X axis
Error Y	Yellow	Indicates switch fault in Y axis
Error Z	Yellow	Indicates switch fault in Z axis

3-7 Mains Input

120VAC 5A for the 5V P.S. through Mains Filter having STD IEC PLUG (GE P/N 2154685)

3-8 Mains Fuse

F1: 7A S.B. 3AG defined as **T7.0AL** (Time delay, 7A, Low breaking current)

3-9 INPUTS

- **Cable Terminals** 6 Terminals ½ inch screws, accepts 6 Gradient Cables from the Amplifiers X/Y/Z.
 - (3 of these terminals, serve as IN/OUT, and therefore will share 2 Cables each).
- **GND.** ¼ inch screw for AWG4 wire connected to the Cabinet Chassis or to **System Gnd.**
- **GP J5:** D-9 Fem.: RS-232 communication to the GIP (or GP).
- **DEBUG TERM J6:** D-9 Male RS-232 communication to a Test Monitor.
- **J1, J2, J3, J4:** Optical connectors (GE 46-307584P3): communication for the future modified GIP(Signals are: **GP DATA, ALARM/READY, GSC DATA, CLK 10**).

3-10 OUTPUTS

Cable Terminals 9 Terminals ½ inch screws for the 6 Coils Cables: WB , ZM, X/Y/Z.
(The other 3 OUT Cables are common to 3 IN Terminals).

3-11 Connectors Details

3-11-1 GP PORT

9 pin Female Sub-D on back panel

PIN #	SIGNAL NAME	DIRECTION	SIGNAL TYPE
2	GIP RxD	INPUT	RS-232
3	GIP TxD	OUTPUT	RS-232
4	GIP DTR	OUTPUT	RS-232
1,5,6,7,8,9	GND	-	GROUND

3-11-2 DEBUG PORT

9 pin Male Sub-D on back panel

PIN #	SIGNAL NAME	DIRECTION	SIGNAL TYPE
2	TXDBG	OUTPUT	RS-232
3	RXDBG	INPUT	RS-232
4	DTR	INPUT	RS-232
1,5,6,7,8,9	GND	POWER	GROUND

3-12 ENVIRONMENT CONDITIONS

OPERATING

Category	Value	Units
Ambient Temperature	5 to 40	°C
Ambient Temperature Change	≤10	°C/hr
Ambient Temperature Gradient	≤10	°C
Humidity (non-condensing)	10 to 90	%RH
Magnetic Field	≤200	Gauss
Altitude	-400 to 11800 (-122 to 3600)	ft (m)

STORAGE, SHIPMENT, AND NON-OPERATING :

Category	Value	Units
Ambient Temperature	-40 to 70	°C
Humidity (non-condensing)	≤95	%RH
Altitude	-400 to 50,000 (-122 to 15000)	ft (m)

4- EXECUTING GRADIENT SWITCH DIAGNOSTICS

Gradient Switch Diagnostics can be executed from the diagnostics menu.

1. Enter the Signa diagnostics menu from the Common Service Desktop
2. Click on the IPG section of the Diags menu. Click again until the word **Datapath** is located under IPG.
3. Click on the box below IPG tests or "Board Level Tests".
4. Click on **MDS**
5. Click on **GSC Communication Test**
6. Run the test.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	August 13, 2001	Ilan Keren	Initial release
1	Feb 22, 2002	Kargard	Added Gradient Switch Diagnostics